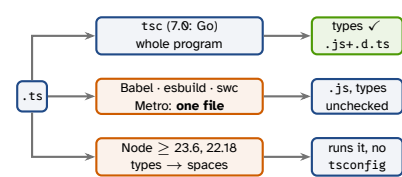
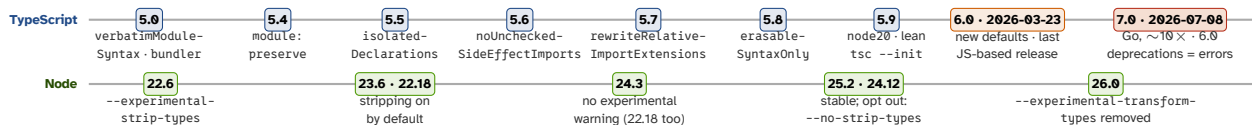


In one line: Types are **erased**: tsc proves consistency *inside* the program, nothing checks the data that enters it. The config decides how strict the proof is, how imports resolve for whoever **loads** the output, and which syntax a per-file transpiler or Node can strip. Since July 2026 tsc is the native Go port (**TypeScript 7**); 6.0 was the last JavaScript-based release.

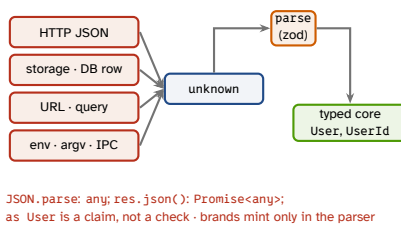
1 who turns .ts into .js?



one file cannot see if {foo} is a type →
isolatedModules · verbatimModuleSyntax ·
erasableSyntaxOnly; types: tsc --noEmitInCI

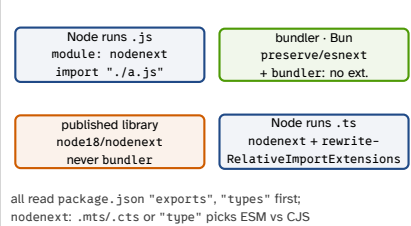


2 erased types → validate at the edge



JSON.parse: any; res.json(): Promise<any>;
as User is a claim, not a check · brands mint only in the parser

3 who loads the output?



Defaults: 5.9 → 6.0 (and 7.0)

option	≤ 5.9	6.0 · 7.0
strict	false	true
module	commonjs	esnext
target	es5	latest spec: es2025
types	every @types/*	[] — list them
rootDir	common dir of inputs	the tsconfig's dir
noUncheckedSideEffectImports	false	true
libReplacement	true	false
stableTypeOrdering	—	opt-in, ~25% slower · 7.0: always

Deprecated in 6.0 (silence: "ignoreDeprecations": "6.0"), **hard errors in 7.0**:
target: es5, downlevelIteration, moduleResolution: node/node10/classic,
module: amd/umd/system/none, outFile, baseUrl (write the prefix into each path entry), esModuleInterop or allowSyntheticDefaultImports: false,
alwaysStrict: false, the module keyword for namespaces, import ... assert (use with), /// <reference no-default-lib>.

Strict, and beyond strict

- strict** = alwaysStrict, strictNullChecks, strictFunctionTypes, strictBindCallApply, strictPropertyInitialization, noImplicitAny, noImplicitThis, useUnknownInCatchVariables (4.4: catch (e) is unknown), strictBuiltinIteratorReturn (5.6: built-in iterators return undefined, not any). New members join on upgrade — pin the TS version.
- Not in strict**: noUncheckedIndexedAccess (4.1, a[i]: T|undefined), exactOptionalPropertyTypes (4.4), noImplicitOverride (4.3), noPropertyAccessFromIndexSignature (4.2), noImplicitReturns, noFallthroughCasesInSwitch, noUnusedLocals/Parameters.
- 5.9 tsc --init** turns on the first two above, plus nodenext, verbatimModuleSyntax, isolatedModules, moduleDetection: force, skipLibCheck, declaration, types: [].
- extends takes an array (5.0; later wins); \${configDir} in path options (5.5) = the directory of the config that is compiled, not of the shared base.
- JS in the program**: allowJs mixes .js in; checkJs or a per-file // @ts-check checks it from JSDoc, // @ts-nocheck opts a file out.

Modules — match the loader

- nodenext**: .mts ESM · .cts CJS · .ts follows the nearest "type". ESM relative imports name the output: ". /mod.js". Default import of CJS = module.exports. Implies esModuleInterop and floating target: esnext; allows require(esm) (5.8, Node 22.12+). Frozen snapshots: node18 (5.8, no require(esm)), node20 (5.9, implies es2023).
- bundler** (5.0): extensionless, reads "exports" with the import condition. **module: preserve** (5.4) implies it, plus esModuleInterop and resolveJsonModule; emits import as written, import x = require() as require. App recipe: esnext+bundler, noEmit, allowImportingTsExtensions (only with noEmit / emitDeclarationOnly), verbatimModuleSyntax; no "type": "module" or .mts in bundler projects.
- Library**: node18, the lowest target you support, strict (or its .d.ts breaks strict users), declaration+declarationMap, rootDir/outDir (else users load your .ts). bundler resolution is "infectious": it accepts imports Node cannot load.
- "exports": TS takes "types" first, then default/runtime conditions; add your own with customConditions. paths only tells TS — tsc never rewrites them, so the bundler / Jest / Node ("imports") needs the same map.

Enums vs unions

Numeric enum = runtime object + reverse map (E[0] is "A"); string enums have none. const enum is inlined: ambient ones break isolatedModules and inline a stale value when the library version changes (handbook: lint them away or preserveConstEnums). Neither survives erasableSyntaxOnly — use an as const object or a literal union.

Erasable TypeScript

- isolatedModules**: errors where one file is not enough — a type re-exported without export type, a namespace in a script file, an ambient const enum.
- verbatimModuleSyntax** (5.0, below): replaced importsNotUsedAsValues and preserveValueImports (gone in 5.5); CJS output needs import x = require().
- erasableSyntaxOnly** (5.8) bans what has runtime meaning: enum, namespaces with code, parameter properties, import = export =.
- Node** swaps types for spaces (lines hold, no source map), ignores tsconfig, needs extensions and import type; no .tsx, no .ts in node_modules. Its docs: panel 3 + verbatimModuleSyntax, erasableSyntaxOnly, target: esnext, noEmit.
- rewriteRelativeImportExtensions** (5.7): ./a.ts → ./a.js — relative literals only; never paths aliases, packages, #/..., computed import().

```
import type { A } from "a";           // erased entirely
import { b, type c } from "bcd";      // -> import { b } from "bcd";
import { type x } from "xyz";         // -> import {} from "xyz";
```

TypeScript 7 — the native compiler

- npm i -D typescript = 7.0.2, still tsc: VS Code 125.7s → 10.6s (11.9×), memory — 6 to ~26%; --checkers N(4), --builders N, --singleThreaded.
- No API in 7.0** (7.1: a new, different one): typed lint rules, webpack loaders, Vue / Svelte / Astro / MDX, Angular templates stay on 6 — @typescript/typescript6 ships tsc6 + the 6.0 API.
- Clean on 6.0 with stableTypeOrdering, no ignoreDeprecations ⇒ should compile identically on 7.0. JSDoc: @enum ignored, values not usable as types.

References, declarations, speed

- composite**: declaration on, rootDir = its dir, every file in include; importers read its .d.ts. tsc -b builds stale projects in order (.tsbuildinfo); since 5.6 past errors (--stopOnBuildErrors).
- isolatedDeclarations** (5.5): exports fully annotated, so any tool emits .d.ts per file, in parallel; --noCheck (5.6) emits without checking.
- Cost**: interfaces over big & (cached); annotated returns; unions past a dozen members compare pairwise. Measure: --extendedDiagnostics, --explainFiles, --generateTrace + @typescript/analyze-trace.

Example — the edges

```
const User = z.object({ id: z.string(), age: z.number() });
type User = z.infer<typeof User>; // one source of truth
async function load(res: Response): Promise<User> {
  return User.parse(await res.json()); // throws ZodError
  declare const tag: unique symbol; // brand: nominal, 0 bytes
  type UserId = string & { readonly [tag]: 'UserId' };
  const Dir = { Up: 'up', Down: 'down' } as const; // not enum
  type Dir = (typeof Dir)[keyof typeof Dir]; // 'up' | 'down'
  // globals.d.ts -- has export: a module, so this augments
  export {}; declare global { var __APP_VERSION__ : string }
  // assets.d.ts -- no import/export: a script, ambient modules
  declare module '*.css' {} // side-effect imports pass
```

Traps

- 6.0 types**: []: describe, process, fs vanish — list "types": ["node", "jest"]. rootDir = .: output lands in dist/src/ unless "rootDir": ". /src".
- declare module "x"** in a file that has any import/export is an augmentation, not a declaration — one stray import turns a shim into a patch.
- @ts-ignore** hides every future error on its line; @ts-expect-error (3.9) fails with Unused @ts-expect-error directive. once fixed. typescript-eslint ban-ts-comment: ignore banned, expect-error needs a ≥3-char reason.